

STAT 301 Midterm Exam

(Time: **60 minutes**, *Thursday, July 23, 2026, 3:45pm–4:45pm*)

Note: 60 minutes for writing your solutions, 15 minutes (4:45pm – 5:00pm) for uploading your solution in a single pdf file to Canvas “Assignment” section (click **“Midterm Solutions upload here”** link).

Name: _____

Student ID: _____

Instructions:

1. Please upload your solutions to Canvas in a **single pdf file**. You have at least two options for writing and uploading your exam solutions: 1) hand-write your solutions on the exam papers in the space provided, scan it, and then upload a single pdf file to Canvas; or 2) write your solutions on (say) OneNote, save it as a single pdf file, and then upload it to Canvas. You may also use other methods, as long as you upload a single pdf file of your solutions to Canvas.

Your solution file name should be in the format of **LastName-FirstName.pdf**, e.g., Wu-Lang.pdf. At the end of the exam, please also hand in the hard copies of your exam/solutions.

2. Please make sure you write **your name and student ID number** on the top of **each page** of your solution file.
3. You must show the **key steps/procedures** of your solutions (unless indicated otherwise). Marking scheme: 70% for correct procedures, and 30% for correct answers. For questions requiring answers in words, it is your responsibility to write clearly (easy to read), or you may lose points.
4. In your solutions, please clearly indicate **the problem number** (e.g., Problem 1 (a)), and circle **your final answer** if a question requiring computation.
5. The midterm exam is closed books/notes. You may bring in **one letter-size sheets of notes** (both sides, written or typed), and a simple calculator. **You are allowed to access Canvas/Internet only at the end of the exam for file uploading.**
6. There are 3 problems, each having several sub-problems, with a total of 47 points.

Problem 1 (16 pts). You wish to study the relationship between salary (y , in thousands of dollars) and years of service x_1 and gender x_2 ($x_2 = 1$ if male and $x_2 = 0$ if female), based on a sample of 100 high school teachers in a large city. You consider the following linear regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + e, \quad (1)$$

where e denotes random error. Based on R function *lm*, you obtained the following estimates (standard errors)

$$\hat{\beta}_0 = 45 \text{ (15)}, \quad \hat{\beta}_1 = 3 \text{ (0.5)}, \quad \hat{\beta}_2 = 4 \text{ (2.5)}, \quad \hat{\beta}_3 = 1 \text{ (0.8)}.$$

(a) (4 pts). Give an estimate of the **difference** in mean salaries of a male teacher with three years of service and a female teacher with three years of service.

(b) (4 pts). A particular **female** teacher with **three** years of service currently has a salary of \$60,000. What is the residual for this observation?

(c) (4 pts). Does the relationship between salary and years of service depend on gender? Answer the question based on a hypothesis testing (with significance level 0.05. Show key steps).

(d) (4 pts). (i) Based on your answer in (c), suggest a simpler model (than model (1)). (ii) If $\hat{\beta}_3 = 4 \text{ (0.8)}$ in Model (1), how might you explain the association between salary and years of service? Answer in less than 30 words.

Problem 2 (15 pts). We want to study the relationship between weight loss (y , in pounds) and time spent on exercise (x_1 , hour/week) and amount of sugar consumption (x_2 , in grams). The following results are R outputs for Model (1): $y = \beta_0 + \beta_1 x_1 + e$ and Model (2): $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e$ respectively based on a sample of 32 observations in a period of one year.

Model (1)			Model (2)		
	estimate	standard error		estimate	standard error
intercept	10.36	8.88	intercept	15.64	9.22
x1	7.96	1.04	x1	9.48	1.54
			x2	-2.57	0.43

(a) (5 pts) (i) Give an interpretation for $\hat{\beta}_1$ in Model (2). (ii) What is the main difference in interpretation for $\hat{\beta}_1$ in Model (1) and $\hat{\beta}_1$ in Model (2)? Limit your answer to each question to no more than 30 words.

(b) (5 pts) (i) Why are the values of $\hat{\beta}_1$ in Model (1) and Model (2) very different? (ii) If the values of $\hat{\beta}_1$ in Model (1) and Model (2) are very close, what does it imply? Answer each question in less than 30 words.

(c) (5 pts) (i) Sketch a possible residual plot as an example where the relationship in Model (1) may **not** be linear. (ii) Sketch a possible residual plot where the variance of the random error may increase with weight loss (so not a constant variance) **and** suggest a common method which may possibly fix the problem. Clearly label the x-axis and y-axis of your plots.

Problem 3 (16 pts). For Problem 2, if the amount of sugar consumption x_2 is recorded in three levels (low, medium, high), and the R output looks like below

Model (1)			Model (3)		
	estimate	standard error		estimate	standard error
intercept	10.36	8.88	intercept	15.64	9.22
x1	7.96	1.54	x1	9.48	1.02
			x2	-1.51	0.38
				-2.01	1.12

(a) (4 pts) Interpret the two estimates corresponding to x_2 in Model (3) respectively in less than 40 words.

(b) (4 pts) Based on the results in Model (3), is the predictor sugar consumption x_2 likely to be significantly associated with weight loss (at 5% level)? Answer in less than 30 words.

(c) (4 pts) Suppose that we are only interested in the relationship between weight loss y and sugar consumption x_2 . Is an analysis of variance (ANOVA) model equivalent to Model (3) and why? Answer in less than 30 words.

(d) (4 pts) (1) Sketch two figures to illustrate that (i) the relationship between weight loss and exercise time does not depend on sugar consumption, and (ii) the relationship between weight loss and exercise time depends on sugar consumption (clearly label the x-axis and y-axis in your plots). (2) Write down the mathematical forms of the two models in (1).